

Geometry
Unit 9
Lesson 11 Practice

Name_____Answer Key_____
Date_____
Period_____

1. When Cheryl was young, she had a twin-size bed. It was 38 inches by 75 inches. What was the base (sleeping) area of the twin bed?
 $A = (38)(75) = 2850 \text{ in}^2$

2. As Cheryl grew taller, she needed a bigger bed. Cheryl purchased a twin XL bed. It was 38 inches by 80 inches. What was the base (sleeping) area of the twin XL bed?
 $A = (38)(80) = 3040 \text{ in}^2$

3. What is the difference in areas of the two beds?
 $3040 - 2850 = 190 \text{ in}^2$

4. With the bed frame, both of Cheryl's beds stood 25 inches high. What was the lateral surface area of the twin bed?
 $(2(38 + 75))(25) = (226)(25) = 5650 \text{ in}^2$

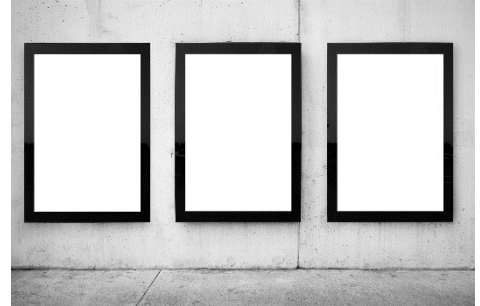
5. What was the lateral surface area of the twin XL bed?
 $(2(38 + 80))(25) = (236)(25) = 5900 \text{ in}^2$

6. Cheryl's grandma made a quilt for the twin XL bed. It was big enough to cover the bed and touch the floor on all four sides. What was the area of the quilt?

$$3040 + 5900 = 8940 \text{ in}^2$$

7. There are three posters on display at the corner store. One is 18 inches by 24 inches, the second is 24 inches by 36 inches, and the third is 27 inches by 40 inches. What is the area of the smallest poster?

$$18 \times 24 = 432 \text{ in}^2$$



8. What is the area of the middle size poster?

$$24 \times 36 = 864 \text{ in}^2$$

9. What is the area of the largest poster?

$$27 \times 40 = 1080 \text{ in}^2$$

10. David designated space on his wall for posters that is 50 inches by 75 inches. Can he fit all 3 posters in his designated space?

It will fit by area, because $50 \times 75 = 3750 \text{ in}^2$, but he will have to cut the posters to make the length and width fit, so no.